Anika Sørensen

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Research interests

How people learn the structure of an environment when feedback is sparse. I combine Bayesian and reinforcement-learning models with EEG and MEG to test how the brain balances exploring against exploiting, with a current focus on adolescence. Studies are pre-registered where possible, and code and data are open.

Education

2019 – 2023 — **PhD, Cognitive Neuroscience**, Nordholm University
Thesis: *Computational models of exploration under uncertainty.* Supervisor: Prof. Elsebeth Krogh. Awarded with distinction.

2016 – 2018 — **MSc, Cognitive Science**, Fjordland University
Graduated with distinction; thesis on reward prediction errors in human EEG.

2013 – 2016 — **BSc, Psychology**, Fjordland University

Appointments

2023 – present — **Postdoctoral Research Fellow**, Centre for Brain and Learning, Nordholm University
Lead two EEG studies on exploration in adolescents (n = 96 and n = 140) and co-supervise two PhD students. Maintain an open-source toolbox for fitting exploration models, used by three labs.

2018 – 2019 — **Research Assistant**, Centre for Brain and Learning
Built the lab's EEG preprocessing pipeline, now used by six projects.

Selected publications

2025 — **Sørensen, A.**, Lindqvist, E., & Bauer, T. Hierarchical priors shape exploration under sparse rewards. *Neural Computation and Behaviour*, 18(3), 211–236.

2024 — **Sørensen, A.** & Kovač, M. Decoding latent task structure from MEG with sparse recurrent models. *Cortex and Computation*, 9(2), e114.

2023 — Larsen, J., **Sørensen, A.**, Okafor, N., & Krogh, E. Reward uncertainty and learning rates in adolescents: a registered report. *Open Psychology Reports*, 4, 55–71.

2022 — **Sørensen, A.**, Krogh, E., & Lindqvist, E. Learning to explore without external rewards. *Proceedings of the Conference on Computational Cognition*, 1402–1411.

2021 — **Sørensen, A.** & Krogh, E. Exploration bonuses in human learning: a meta-analytic review. *Neural Computation and Behaviour*, 15(1), 33–58.

In preparation — **Sørensen, A.** et al. Exploration and sleep in adolescents: a pre-registered EEG study.

Grants and fellowships

2024 – 2026	**Early Career Research Grant**, Nordic Foundation for Brain Research DKK 1.2 million as principal investigator, for the adolescent exploration studies.
2022	**Student travel award**, Conference on Computational Cognition
2019 – 2023	**PhD Fellowship**, Nordholm Graduate School

Teaching and supervision

2024 – present	**Lecturer**, Computational Modelling of Behaviour (MSc) About 40 students a year; rebuilt the course around reproducible notebooks.
2023 – present	**Co-supervisor** of two PhD students and five MSc theses
2020 – 2023	**Teaching assistant**, Research Methods (BSc) Four cohorts, about 180 students.

Invited talks

2025	Why do adolescents explore more? *Brain and Learning Symposium, Aarhus*
2024	Learning structure from sparse feedback. *Winter School on Computational Cognition, Oslo*
2023	Models of exploration in the developing brain. *Invited seminar, Fjordland University*

Conference presentations

2025	Talk, Conference on Computational Cognition: hierarchical priors and exploration.
2024	Poster, Organization for Human Brain Mapping Nordic meeting: MEG decoding of task structure.
2022	Talk, Conference on Computational Cognition: learning to explore without rewards.

Service and open science

Reviewing	*Neural Computation and Behaviour*, *Open Psychology Reports*, *Cortex and Computation*
2021 – present	Organiser, Nordholm Methods Reading Group (monthly, 25 regular attendees)
Open science	All studies pre-registered on OSF; analysis code and anonymised data released with each paper.

Methods and tools

Methods	EEG, MEG, eye tracking; Bayesian and reinforcement-learning models; mixed-effects statistics
Software	Python (NumPy, PyTorch, JAX), R, MATLAB, MNE-Python, Stan, Git
Languages	Danish (native), English (fluent), German (B2)

References available on request